VACUUM

Editor-in-Chief

Lars G Hultman, *Thin Film Division, Department of Physics, Linköping University, S-58183 Linköping, Sweden, E-mail: lars.hultman@liu.se*

Editor

Per Eklund, *Dept. of Physics, Chemistry, and Biology (IFM), Linköping University SE-581 83 Linköping, Sweden, E-mail: per.eklund@liu.se*

Associate Editors

Oleg B Malyshev, *STFC, Daresbury Laboratory, Daresbury, Warrington, Cheshire WA4 4AD UK, E-mail: oleg.malyshev@stfc.ac.uk*

Paul H Mayrhofer, *Institute of Materials Science and Technology, Vienna University of Technology, A-1040 Vienna, Austria, E-mail: paul.mayrhofer@tuwien.ac.at*

Luigia Sabbatini, *Chemistry Department, University of Bari "Aldo Moro", I-70126 Bari, Italy, E-mail: luigia.sabbatini@uniba.it*

VACUUM UNITS TO BE USED IN 'VACUUM'

Pressure: Authors are asked to use bar (bar), millibar (mbar) or the pascal (Pa).

The SI unit of pressure is the pascal (Pa), the special name for the newton per square metre (N m^{-2}). The standard atmosphere (atm) is defined in ISO 31/111-1978 (E) as exactly 101.325 kPa. The standard atmosphere should only be used to define a reference environment and not as a pressure unit. The 'bar' has become widely used as a unit of pressure in vacuum measurement.

1 bar = 10^5 Pa exactly

1 millibar (mbar) = 100 Pa exactly.

Other units of pressure are sometimes to be found in published literature but their use is depercated in ISO 31/111-1978(E) and in some countries their use is unlawful for certain purposes. A table of conversion factors for such units is given below to allow authors to change their units to pascals or millibars.

The pascal can be combined with other SI units to give the units for other quantities commonly measured in vacuum science and technology. Some examples of these quantities are given below.

Pump speed: Authors are asked to use cubic metre per second ($m^3\ s^{-1}$), cubic metre per hour ($m^3\ h^{-1}$), or, litre per second ($l\ s^{-1}$).

The SI unit for volumetric rate of flow such as is used when expressing the speed of a vacuum pump is cubic metre per second ($m^3\ s^{-1}$). The speed of most mechanical pumps is expressed in terms of cubic metre per hour ($m^3\ h^{-1}$).

1 $m^3\ h^{-1}$ = 1/3600 $m^3\ s^{-1}$ (0.000278 $m^3\ s^{-1}$).

The speed of most high vacuum pumps such as diffusion and turbomolecular pumps is expressed in terms of liter per second ($l\ s^{-1}$).

1 $l\ s^{-1}$ = $10^{-3}\ m^{-3}\ s^{-1}$.

Leak rate: Authors are asked to use pascal metre cube per second (Pa $m^3\ s^{-1}$) or millibar litre per second (mbar $l\ s^{-1}$).

The SI unit for leak rate is pascal metre cube per second (Pa $m^3\ s^{-1}$). Other commonly used units of leak rate with their conversion to the SI unit are given in the table.

Outgassing rate: Authors are asked to use pascal metre cube per second per square metre (Pa $m^3\ s^{-1}\ m^{-2}$) or millibar litre per second per square centimetre (mbar $l\ s^{-1}\ cm^{-2}$).

The SI unit for outgassing rate from surfaces is pascal metre cube per second per square metre (Pa $m^3\ s^{-1}\ m^{-2}$). Other commonly used units of outgassing rate with their conversion to the SI unit are given in the table.

Table of conversion factors

(1) Quantity measured in	(2) Multiply by	(3) Resulting number in	(4) Multiply column (3) by	(5) Resulting number in
torr	133.322	Pa	0.01	mbar
mm Hg	133.322	Pa	0.01	mbar
micron	0.133322	Pa	0.01	mbar
technical atmospheres ($kgf\ cm^{-2}$)	98.0665	kPa	10	mbar
$m^{-3}\ h^{-1}$	0.000278	$m^3\ s^{-1}$		
ls^{-1}	0.001	$m^3\ s^{-1}$		
torr $l\ s^{-1}$	0.133322	Pa $m^3\ s^{-1}$	10	mbar $l\ s^{-1}$
standard atmosphere ($cm^3\ s^{-1}$)	0.101325	Pa $m^3\ s^{-1}$	10	mbar $l\ s^{-1}$
torr $l\ s^{-1}\ cm^{-2}$	1333.22	Pa $m^3\ s^{-1}\ m^{-2}$	0.001	mbar $l\ s^{-1}\ cm^{-2}$

EDITORIAL ADVISORY BOARD

G. Abadias, *Université de Poitiers, Institut Pprime (UPR 3346), Dpt. Physique et Mécanique des Matériaux SP2MI, Téléport 2, Bd. Marie et Pierre Curie 86962 Chasseneuil-Futuroscope, France*

M. Anderle, *Autonomous province of Trento, Knowledge Department, Via Jacopo Aconcio, 5, 38122 Trento, Italy*

P. Ayala, *University of Vienna, Faculty of Physics, Vienna, Austria*

K. Baba, *Industrial Technology Center of Nagasaki, Omura, Nagasaki 89560026, Japan*

S. Baba, *Faculty of Engineering, Seiki University, Musashino-shi, Kitsushouji Kitamachi, 3-3-1, Japan*

S. Baragetti, *University of Bergamo, viale Marconi 5, 24044 Dalmine BG, Italy*

L. Bardos, *Department of Engineering Sciences, Division of Electricity, Uppsala University, Box 534, S-751 21 Uppsala, Sweden*

F. Berto, *Norwegian University of Science and Technology, 7491 Trondheim, Norway*

H. Biederman, *Charles University, Faculty of Mathematics and Physics, V Holešovickách 2, 180 00 Praha 8, Czech Republic*

W. Dai, *Pennsylvania State University, 332A Davey Laboratory University Park, PA 16802, USA*

J.L. de Segovia, *Instituto de Ciencia de materiales de Madrid, ICMM.CSIC, 28049 Cantoblanco (Madrid), Spain*

Y. Du, *State Key Lab of Powder Metallurgy, Central South University, Changsha, Hunan, 410083, P. R. China*

R.G. Elliman, *Research School of Physical Sciences and Engineering, Australian National University, Canberra, ACT 0200, Australia*

A. Erdemir, *Argonne National Laboratory, ET/212, 9700 South Cass Avenue, Argone, IL 60439, USA*

A. Goodyear, *The open University, Faculty of Technology, Materials Engineering Department, Walton Hall, Milton Keynes MK7 6AA, UK*

J.-H. Huang, *Department of Engineering and System Science, National Tsing Hua University, 101 Kuang Fu Road, Section 2, Hsinchu, Taiwan*

R.E. Hurley, *School of Electrical Engineering and Computer Science, Northern Ireland Semiconductor Research Centre, The Queen's University of Belfast, Stranmillis Road, Belfast, BT9 5AH, Northern Ireland, UK*

A. Ignatiev, *Center for Advanced Materials, University of Houston, Houston, TX 77204-5004, USA*

K. Jousten, *Physikalisch-Technische Bundesanstalt, Section Vacuum Metrology, Abbestr. 2-12, 10587 Berlin, Germany*

D.S. Karpuzov, *Department of Chemistry, University of Alberta, Edmonton, Alberta T6G1H9, Canada*

R. Kersevan, *CERN Vacuum Surfaces and Coatings Group, Mailbox M28800 1211 Geneva 23, Switzerland*

S. Kodambaka, *University of California at Los Angeles (UCLA), Dept. of Materials Science & Engineering, 3121B Engineering V Los Angeles, CA 90095-1595, USA*

M.K. Lei, *Dalian University of Technology, School of Materials Science and Engineering, Surface Engineering Laboratory, 116024 Dalian, China*

K. Ludwig, *Boston University, Dept. of Physics, 590 Commonwealth Avenue, Boston, MA 02215, USA*

E. Mazzotta, *Università del Salento, Dipartimento di Scienze e Tecnologie Biologiche ed Ambientali, Lecce, Italy*

R. Miles, *Newcastle Photovoltaic Applications Centre, University of Northumbria, Ellison Place, Newcastle-upon-Tyne, NE1 8ST, UK*

Y. Mishra, *Institute for Materials Science, Kiel University, Kaiserstr. 2, D-24143 Kiel, Germany*

A.Z. Moshfegh, *Professor & Chairman, Dept. of Physics, Sharif Univ. of Technology, P.O. Box 11155-9161, Azadi Avenue, Tehran, Iran*

D. Music, *Rheinisch-Westfälische Technische Hochschule Aachen (RWTH), Materials Chemistry, Kopernikusstrasse 16, D-52056 Aachen, Germany*

J. Musil, *Faculty of Applied Sciences, University of West Bohemia, P.O. Box 314, 306 14 Plzen, Czech Republic*

V. Nagirnyi, *Institute of Physics, University of Tartu, Riia 142, 51014 Tartu, Estonia*

P. Pedersen, *Department of Physics, Chemistry and Biology, Linköping University, Sweden*

I. Petrov, *Materials Science Department, The Coordinated Science Lab and Materials Research Lab, University of Illinois, 1101 W Springfield Ave, Urbana, IL 61801, USA*

H. Pouraliakbar, *WorldTech Scientific Research Center (WT-SRC), Tehran, Iran*

A. Prakash, *Intersil, A Renesas Company, 3013, White ash trail, Orlando, FL-32826, USA*

C. Rego, *School of Science and the Environment, Manchester Metropolitan University, Chester Street, Manchester, M1 5GD, UK*

F. Richter, *Technical University Chemnitz, Solid State Physics, Reichenhainer Str. 70, D-09107 Chemnitz, Germany*

A. Rossi, *Dipartimento di Scienze Chimiche e Geologiche, Università di Cagliari, Cagliari. Italy*

Y. Saito, *Accelerator Division, KEK-High Energy Accelerator Research Organization Oho 1-1, Tsukuba, Ibaraki, 305-0801, Japan*

J. Setina, Pressure Metrology Lab, *Institute of Metals and Technology, Ljubljana SI-1000, Slovenia*

F. Sharipov, *Departamento de Fisica, Universidade Federal do Parana, Caixa Postal 19044, 81531-990 Curitiba, Brazil*

L. Siller, *The School of Chemical Engineering and Advanced Materials, University of Newcastle upon Tyne, Newcastle upon Tyne NE1 7RU, UK*

C.Q. Sun, *School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798*

Y. Tanimoto, *High Energy accelerator Research Organization, KEK, Japan*

H.R. -Tragenreif, *TU Wien University, Vienna, Austria*

D. Valougeorgis, *University of Thessaly, Department of Mechanical Engineering, Leoforos Athinon, Pedion Areos, 38334 Volos, Greece*

H. Wang, *Dept. of Electrical and Computer Engineering, Materials Science and Engineering Program, Texas A & M University, College Station, TX 77843-3128, USA*

M.-S. Wong, *Institute of Materials Science and Engineering, National Dong Hwa University, Shoufeng, Hualien, Taiwan*

V.E. Yurasova, *Faculty of Physics, Moscow State University, Moscow 117324, Russia*

W.T. Zheng, *Department of Materials Science, Jilin University, Changchun 130012, PR China*

Y. Zhou, *Shenzhen University, Shenzhen, China*

We would like to thank all previous Editorial Board members and reviewers for their valuable time and help with production of this journal.